

Meta-Structure

IB 514: Scientific Writing

Audience Identification

Meta-
Structure

Audience &
Journal

Organization
and Flow

Journal Article
Types

Content
Strategies

Additional
Sand-Jensen
Principles

Final
Thoughts

- Know your target audience:
 - General scientific audience
 - Specialists vs. interdisciplinary researchers
 - Theoreticians and/or empiricists
- Audience shapes decisions about:
 - Terminology and background explanation
 - Context and significance framing
 - Level of technical detail
 - Visual presentation choices (e.g., conceptual figs)
- Tailor writing to match audience expertise and interests
- Consider what audience knows vs. needs to know

Journal Selection

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Thoughts

- Identify target journals *before* you start writing
- Rank potential journals:
 - Aspirational journals (high impact, broad audience)
 - Realistic journals (good fit for topic and scope)
 - Backup journals (alternative options if rejected)
- Consider:
 - Similarity of published articles to your work
 - Journal scope and audience
 - Open access policies (and costs)
 - Impact factor and prestige (with caution)
- If you're not confident
 - Check target journal requirements for alignment
 - Review recent issues

Signposting

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Thoughts

- Guide readers through your argument with effective signposting
- Use transitions and connectives:
 - Topic sentences that signal direction
 - Transitional phrases linking paragraphs and sections
 - Explicit statements of relationships between ideas
- Employ structural markers:
 - Clear section headings and subheadings
 - Summary sentences to recap key points
- Help readers anticipate and understand:
 - What comes next
 - How ideas connect
 - Why information matters

Structural Coherence

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Thoughts

- Maintain logical flow throughout the paper
- Structure provides the “roadmap” for readers
- Key aspects:
 - Logical progression of ideas within sections
 - Clear connections between sections
 - Motivation established before presenting methods
 - Results presented before interpretation
 - Conclusions grounded in findings
- Ensure each section connects to and builds on previous

- 1 **Methods** first (most structurally constrained).
- 2 **Results** second (figure-driven; claim-evidence alignment).
- 3 **Introduction** after results clarify true contribution.
- 4 **Discussion** last to interpret, generalize, and situate.

Introduction (Why?)

- Move from broad problem to specific knowledge gap.
- Critically evaluate literature (don't summarize passively).
- End with explicit research questions, hypotheses, or key finding.

Methods (How?)

- Provide enough detail for replication.
- Use clear subheadings aligned with research questions.
- Justify key analytical decisions.

Results (What?)

- Organize around figures/tables.
- Lead each subsection with the main finding, align with Methods sections
- Avoid interpretation beyond minimal context.

Discussion (So what?)

- Interpret findings relative to hypotheses.
- Compare with prior literature.
- Address limitations and articulate broader implications.

Methods-at-the-End Papers

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Design a Figure-Driven Narrative

- Determine the logical sequence of core results.
- Ensure each figure advances the central claim.
- Remove peripheral analyses from main text.

Front-Load Essential Context

- Provide minimal methodological orientation in Results.
- Define key variables and datasets before presenting outcomes.

Write Results as a Coherent Argument

- Each subsection should answer a focused question.
- Maintain conceptual momentum between figures.

Develop Detailed Methods Section at End

- Provide full reproducibility detail.
- Use clear subsections for sampling, measurements, analyses.
- Anticipate reviewer questions and address proactively.

Key: Requires exceptionally clear storytelling

Literature Reviews

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- **Define scope precisely**
 - What question is the review answering?
 - What time period, disciplines, or methodologies are included?
- **Develop a transparent search strategy**
 - Databases used, search terms, inclusion/exclusion criteria.
 - Consider a flow diagram for study selection.
- **Organize by conceptual structure (not chronology)**
 - Thematic clusters
 - Competing hypotheses
 - Methodological approaches
- **Synthesize within sections**
 - Compare studies directly.
 - Highlight agreements, disagreements, and methodological drivers.
- **Conclude with implications**
 - Identify unresolved questions.
 - Clarify future research directions.

Synthesis

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Thoughts

- **Begin with a conceptual problem**
 - What tension, paradox, or fragmentation exists?
- **Abstract upward**
 - Move from specific findings to higher-level principles.
- **Integrate across domains**
 - Combine theories, datasets, or methods not typically linked.
- **Develop a unifying framework**
 - Conceptual diagram or model.
 - Clear definitions of key constructs.
- **Demonstrate explanatory gain**
 - Show how the synthesis resolves contradictions or generates predictions.

Data Papers

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Thoughts

- Clarify dataset purpose and potential reuse scenarios.
- Provide full documentation:
 - Sampling design, instrumentation, preprocessing steps.
 - File structure and variable-level metadata.
- Demonstrate quality control
 - Validation checks, error screening, uncertainty description.
- Include example use cases
 - Illustrative analyses showing dataset functionality.

Methods, Results & Supplementary Materials

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Methods and Results

- Sufficient detail for general understanding & reproducibility
- Balance completeness with readability
- Use parallel subheadings to organize

Supplementary Materials

- Use strategically to keep main text focused and concise
- Main text must remain complete and independent
- Include:
 - Extended methods, secondary or alternative analyses
 - Statistical analysis details or results tables
 - Details and results that are not essential to main narrative

Integration of Equations

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- Introduce equations before presenting them
- Interpret equations after presenting them
- Equations (symbols) are words in a sentence, not standalone objects (Use appropriate grammar and punctuation)
- Guidelines:
 - Define all variables and parameters
 - Explain the logic and significance of equations to broad audience
 - Follow standard (or explain) mathematical notation
- Keep equations in the main text only if essential to narrative¹

¹See Fawcett & Higginson, *PNAS* 2012 and responding commentaries

Designing an Effective Title

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Most-read part — determines if reader even looks at paper!

- **Be specific:** state the finding, not the topic
- **Aim for 8–14 words:** enough to be precise, short enough to scan
- **Front-load key terms** for visibility
Avoid: “A study of. . .”, “The role of. . .”, “Effects of X on Y”
- **Consider style:** declarative sentence, question, or noun phrase
A noun phrase has no main verb (essentially a sophisticated label).
“Temperature dependence of...” names phenomenon without asserting anything about it.
- **Catchy, clear, concise²**

Weak	<i>A Study of predator foraging behavior in an aquatic system</i>
Stronger	<i>Temperature mediates Type II functional responses in a dragonfly–Daphnia predator–prey system</i>
Strongest	<i>Temperature dependence of attack rates but not handling times in a dragonfly–Daphnia predator–prey system</i>

²See Singh Chawla (2025)

Structuring an Effective Abstract

Meta-Structure

1–2 sentences Broad introduction to the field, accessible to any scientist

2–3 sentences Detailed background for scientists in related disciplines

1 sentence The specific problem this study addresses

1 sentence The main result — use *“here we show”* or equivalent

2–3 sentences What the result reveals relative to prior understanding; what it adds to existing knowledge

1–2 sentences Broader context and implications

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Final Thoughts

Key Principles³

- Self-contained: interpretable without access to the full paper
- Concise: limit to essential details and key message
- Include quantitative results where possible (e.g., *“attack rates declined by 43% at 10° C relative to 20° C, with handling time showing no change.”*)
- State implications explicitly: do not leave the reader to infer them

³See *Nature-Summary-Paragraph.pdf*

Write Concisely

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Brevity signals confidence in argument — length can signal the opposite

Principles

- **One idea per sentence;** one purpose per paragraph
- **Cut throat-clearing:** 1st sentence can often be deleted
- **Prefer the shorter word:** “use” not “utilize”; “show” not “demonstrate”
- **Avoid redundancy:** “completely eliminate”, “end result”, “past history”

A Useful Test

After drafting, ask: *does every sentence earn its place?* If a sentence neither advances the argument nor provides essential context, delete it.

Verbose	<i>“It is important to note that the results of the analysis that was conducted clearly demonstrate that. . .”</i>
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Concise	<i>“The analysis shows that. . .”</i>
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Discussion Writing: Implications and Speculation

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The Discussion is Where Science Moves Forward

Stating only what you found is not enough. Readers need to know what your findings *mean* and where they lead.

A strong discussion . . .

- **Answers the question posed in the Introduction** — return explicitly to hypothesis or objective
- **Compares findings to prior work** — not just “*consistent with*” but *why* agreement or disagreement matters
- **States implications directly** — do not leave the reader to infer
- **Licenses informed speculation** — clearly flagged (“*we suggest. . .*”, “*one possibility is. . .*”) but not avoided
- **Identifies limitations honestly** without undermining the contribution
- **Opens new questions** — the best papers leave a field more curious than they found it

Watson & Crick (1953): “*It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.*” Had they omitted this implication, acceptance of the model would have been substantially delayed.

Figure-Driven Storytelling

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Figures Are Not Decoration

A well-designed figure conveys structure, mechanism, or pattern that would require thousands of words to describe — and is more memorable.

- **Plan figures before writing:** if you can sketch your story as a sequence of figures, your narrative structure is sound
- **Each figure should make one clear claim** — if it requires extensive caption explanation, simplify
- **Figures and text must be mutually reinforcing**, not redundant — do not describe in prose what is already visible in the figure
- **Captions should be self-contained:** a reader should understand the figure without reading the main text
- **Order figures to mirror narrative arc** — the figure sequence is a parallel story

Use figures creatively and purposefully

Conceptual diagrams	Hypotheses, frameworks, food webs
Data figures	Results; patterns across treatments or gradients
System figures	Study site, organism, or experimental design
Synthesis figures	Meta-analytic summaries; effect size plots

Managing Jargon and Terminology

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Jargon is a double-edged sword

Exists to enable precision, not to signal membership. Overuse narrows audience, obscures weak reasoning, and slows reader.

- **Define terms on first use** — even for a specialist audience
- **Use abbreviations sparingly** — only for terms used $\gtrsim 5$ times
- **Calibrate to audience** — what is jargon for general reader may be essential shorthand for specialist
- **Be especially careful in:** the Abstract (broadest readership), Introduction (context-setting), and Discussion (implications must be accessible)

<i>Jargon-heavy</i>	<i>"Spatiotemporal heterogeneity in resource availability mediates trophic cascade attenuation across productivity gradients."</i>
<i>Clearer</i>	<i>"Patchy resource availability weakens top-down control by predators in more productive systems."</i>

Final Thoughts

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Thoughts

- **Meta-structure is the skeleton** of your paper — it supports and organizes all content
- **Audience and journal** shape meta-structure — tailor to fit the norms and expectations of your target
- **Structural coherence** guides readers through your argument — signposting and logical flow are essential